

CSE 331

AUTOMATA AND COMPUTABILITY

Building DFA

Spring 2026

Building DFA



Typical questions

Define a DFA which recognizes the given language L .

or

Prove that the (given) language L is regular.

Building DFA



Example

Define a DFA which recognizes

$$L_1 = \{w \mid \text{all strings containing at least three 0s.}\}$$

Building DFA



Example

Define a DFA which recognizes

$$L_2 = \{w \mid \text{the length of } w \text{ is one more than a multiple of } 3\}$$

Building DFA



Example

Define a DFA which recognizes

$$L_3 = \{w \mid w \text{ when interpreted as a binary number, is divisible by } 3\}$$

Building DFA



Example

Define a DFA which recognizes

$L_4 = \{w \mid \text{all characters are repeated at least once wherever they appear in the string}\}$

Building DFA



Example

Define a DFA which recognizes

$L_5 = \{w \mid w \text{ contains an equal number of occurrences of the substrings } 01 \text{ and } 10 \}$

Building DFA



Example

Define a DFA which recognizes

$$L_6 = \{w \mid 01 \text{ appears even number of times in } w \text{ as a substring}\}$$

Building DFA



Example

Define a DFA which recognizes

$L_7 = \{w \mid \text{all strings in which every run of 0s has length at least 3}\}$

Building DFA



Example

Define a DFA which recognizes

$L_8 = \{w \mid \text{all strings } w \text{ such that in every prefix of } w, \text{ the number of 0s and 1s differ by at most } 1\}$

Building DFA



Example

Define a DFA which recognizes

$L_g = \{w \mid \text{all strings containing at least two 0s and at least one 1.}\}$



Building DFA



Remember

States are our only (computer) memory.

Design and pick states with specific roles / tasks in mind.

"Have not seen any of desired pattern yet"

"Trap state"